

A Feature Selection Method Based on Information Gain and Genetic Algorithm

Shang Lei

Department of Information Science and Technology
Shandong University of Political Science and Law
Jinan, China
leilishang@163.com

Abstract—with the rapid development of the Computer Science and Technology, It has become a major problem for the users that how to quickly find useful or needed information. Text categorization can help people to solve this question. The feature selection method has become one of the most critical techniques in the field of the text automatic categorization. A new method of the text feature selection based on Information Gain and Genetic Algorithm is proposed in this paper. This method chooses the feature based on information gain with the frequency of items. Meanwhile, for the information filtering systems, this method has been improved fitness function to fully consider the characteristics of weight, text and vector similarity dimension, etc. The experiment has proved that the method can reduce the dimension of text vector and improve the precision of text classification.

Keywords—Information Gain; Genetic Algorithm; Feature Selection; information filtering

I. INTRODUCTION

With the rapid development of the Computer Science and Technology, the electronic information is increasing and exploding. It has become a major problem for the users that how to quickly find useful or needed information in the vast informations. Information Retrieval technology can help people to partially solve this problem. Web Search Engine is a typical application of the Information Retrieval technology. In order to get the most accurate information efficiently and effectively from a range of diversities, text expression and auto-categorization technology has drawn extensive attractions from the public for the difference it can make. Recommendation system based on the use of recommended techniques can be divided into two types: rule-based systems and information filtering system. Information filtering systems can be divided into content-based filtering systems and collaborative filtering systems. Using text feature to present text information in vocabulary level is an effective way for text categorization. [1]

Text information filtering mainly involves three aspects: a) the representation and pre-treatment of text information, b) the establishment of the user profile, and c) matching the text with the user profile. Among them, the representation and pre-treatment is one of the most basic processes in information filtering. Currently the vector space model, which represent the word in the text as feature items, is mainly used. However, in this form the number of the text vector dimension is very large.

It is difficult to improve the speed and accuracy of the text automatic classification. Therefore, it is necessary to take the original feature space dimensionality reduction process [2] before the classification. This process is named feature selection, and selects the candidate including the most effective and the greatest contribution of the characteristics of text classification from a group of the features. In fact, feature selection process is actually a process of optimization. In this paper, the text feature selection method based on Information Gain and Genetic Algorithm is proposed, which searches in the feature space using the idea of evolutionary computation, in order to find the optimal feature subset. In Section 2, some related works are given. The key technologies are proposed in Section 3. The whole algorithm of this new feature selection method is proposed in Section 4. In Section 5, we give the experiment and analysis of the results. And in the last section, we conclude our works and point out the works we will do in the future.

II. RELATED WORKS

At present, there are several common methods for the feature selection, such as document frequency threshold method, information gain method, distribution method, mutual information entropy method and the latent semantic indexing method based on singular value decomposition. These feature selection methods have proved all have some limitations [3].

Genetic Algorithm is a random search and optimal method which based on biology natural evolutionary and genetic mechanism. It followed the process of the evolution and genetic of biology, started from a set of feasible initial population, adopt the operation of selection crossover and mutation and so on, and gradually approach the optimal solution to the problem. It was first proposed by Professor J. Holland of Michigan University in United States in 1975. It had been applied and developed in the areas of pattern recognition, machine learning, neural networks, combinatorial optimization and so on. However, traditional genetic algorithms, exist a lot of problems, large storage capacity calculation, can not guarantee to convergence to the global optimum [4], the best chromosome loose in group and premature convergence in the process of the evolution of biology and so on, to solve these problems, researchers propose a new Genetic Algorithms—Quantum Genetic Algorithm based on quantum computation combined with genetic algorithms. Genetic Algorithm is an

optimization method which based on the natural selection and genetic selection of the biological evolution of drawing.

III. KEY TECHNOLOGIES OF FEATURE SELECTION METHOD BASED ON INFORMATION GAIN AND GENETIC ALGORITHM

A. Information Gain

Information Gain (IG) is an entropy-based feature evaluation method, widely used in the field of machine learning. As Information Gain is used in feature selection, it is defined as the amount of information provided by the feature items for the text category. Information gain is calculated by how much of a term can be used for classification of information, in order to measure the importance of lexical items for the classification. The formula of the information gain is shown below.

$$\begin{aligned} G(D, t) = & -\sum_{i=1}^m P(C_i) \log P(C_i) \\ & + P(t) \sum_{i=1}^m P(C_i|t) \log P(C_i|t) \\ & + P(\bar{t}) \sum_{i=1}^m P(C_i|\bar{t}) \log P(C_i|\bar{t}) \end{aligned} \quad (1)$$

In Formula 1, C is a set of document collection, in which there is not the feature t . The value of $G(D, t)$ is greater; t is more useful for the classification for C . This t should be selected. If the greater value of $G(D, t)$ is wanted, it should make the value of $P(t)$ and $P(\bar{t})$ smaller.

B. Genetic Coding

On the feature set $F = \{f_1, f_2, \dots, f_N\}$, each of the genetic characteristics is encoded by use of binary, a N code length strings of binary code is gives: $h = h_1 h_2 \dots h_N$, the encoding feature set of strings that made a choice, in which $h_i = 1$ that the characteristics of the first i was chosen, all the features of selected form a feature subset.

$H = \{h_1, h_2, \dots, h_N \mid h_i \in \{0, 1\}, i = 1, 2, \dots, N\}$ for all the characteristics of a subset of the collection, known as the individual space. The size of individual space as follows:

$$C_N^1 + C_N^2 + \dots + C_N^N = 2^N - 1 \quad (2)$$

C. Fitness function

Based on our analysis of the characteristics of text filtering, we consider the fitness function in the following points:

First, the text can be distinguished from other categories. We can measure the similarity to be a better individual for the same goal in that the population should be more similar to the individual, while a poor individual has a little similar to the individual, the first i individuals of type of similarity can be calculated as follows:

$$\text{simdeg}(x_i) = \frac{|\text{simnum}(x_i)|}{m} \quad (3)$$

$$\text{simnum}(x_i) = \{set(x_i) \mid i \neq j \wedge \text{sim}(x_i, x_j) \geq \theta\} \quad (4)$$

Where, $\text{simnum}(x_i)$ is the similar between x_i , and its number of individuals is over θ . θ is the set of similarity threshold. $\text{sim}(x_i, x_j)$ is the individual of x_i and x_j . The similarity is calculated by cosine of two vectors.

Feature vector dimension should be less. Text feature selection goal is to use a small number of characteristics to achieve the same or better classification results, therefore, the same ability to express the individual, It is better the number of its characteristics is less, reflected in the individual coding is "0", the number which is the more the better, with $|zeros|$ that the individual number of 0.

Therefore, the fitness function can be expressed as follow.

$$\begin{aligned} f = & \alpha \times \bar{\omega}_i + \beta \times \text{simdeg}(x_i) + \gamma \times \frac{|zeros|}{n}, \\ & \alpha + \beta + \gamma = 1 \end{aligned} \quad (5)$$

D. Selection, crossover and mutation

The uses of genetic algorithms for selecting operator groups operate the survival of the fittest individuals. Commonly used method of operation with a lot of choices, there are roulette wheel selection, elitist selection, non-random selection of playback. Choices randomly selected from the population a certain number of individuals, and then the individual will be selected as the best parent individual. Repeat this process for the completion of individual choice. This method can avoid premature convergence to some extent, is also very strong randomness.

Crossover is the populations of an important means of generate a new individual, overlapping many ways, there is cross-linear, discrete crossover and cross the middle. Crossover operation mimics the natural evolution of organisms in the process of genetic recombination; genetic algorithm will be through the generation of cross-operation of the fine individual genes to the next generation. The purpose of cross-operation is good for new individuals to maintain population diversity.

Mutation operation is to mimic natural evolution of organisms chromosome gene mutation phenomenon occurred. It is an important new means to the individual so as to maintain the diversity of the population. The purpose of mutation operation is two-fold: First, the genetic algorithm to improve the local search capabilities and the other is to maintain the population diversity to prevent premature.

IV. ALGORITHM PROCESSES

In this paper, the characteristics of the text of feature selection are given to the Feature Selection Method based on the Information Gain and Genetic Algorithm for text specific steps:

1) First of all, for the original feature set, using the MI algorithm to improve the initial screening, to remove redundancy and noise characteristics, the formation of feature subset.

2) Population initialization $Q(t)$. Calculation of initial population $Q(t)$ of each individual, to be a group of state like below.

$$P(t_0) = \{p_1^{t_0}, p_2^{t_0}, \dots, p_n^{t_0}\} \quad (6)$$

In which $p_n^{t_0}$ is for the t_0 generation population of solutions of the first j individual, the form of binary m length of string.

3) The use of selection, crossover, mutation operator to generate offspring, offspring of the new value for assessment of fitness.

4) Record the best individual state, its fitness value and compare the current target value, if greater than the current target, then the new optimal solution as the next iteration of the target, otherwise, maintain the current the target unchanged.

5) The end of algorithm.

V. EXPERMENTS

In the experiment, we use the IG and Genetic Algorithm to improve the training of the characteristics of Collection items weight basis, and on this basis the implementation of genetic algorithm training template, so as to arrive at the issue of space on behalf of the Optimal Feature Subset. The parameters set as follows: 80 from the initial population size, mutation probability $P_m=0.06$, in the fitness function, $\alpha=0.1$, $\beta=0.6$, $\gamma=0.3$, set the terms for terminating a genetic adjacent generations to adapt to the average margin of 0.0001 degrees, the greatest genetic algebra for 1000.

This experiment use the training text of experiment were from the People's Network, a total of 2000 articles of six categories, including 1500 articles as a training sample and 500 articles as a testing text, the experimental contain sub-feature vector access stage and verify the correctness stage. At the same time, in order to get a better effectiveness of the algorithm, we have introduced the rate of about reduction E and the accuracy rate F to evaluate the two indicators. And E and F are defined as follow.

$$E = (1 - \frac{N_r}{N}) \times 100\% \quad (7)$$

Where N is the original feature dimension, N_r is the dimension after reducing the eigenvector.

$$F = \frac{n_c}{n} \times 100\% \quad (8)$$

Where n_c is the number of correct text reducing, and n is the total number of text.

In the text filtering step, we have done the necessary pre-processing of the text, such as to deal with the low-frequency words, to stop words, punctuation, etc., and then done a number of experiments with Naive Bayesian to text classifier. We selected five records in the whole results of these experiments, as shown in table 1:

TABLE I. THE EXPERIMENTAL RESULTS

Serial No.	The Results of the Experiment			
	The number of text correct classification	The number of features after reduction	Reduction rate E	Accuracy rate F
1	475	2312	69.32%	87.31%
2	434	1923	70.42%	83.20%
3	453	1820	71.23%	83.73%
4	457	2103	69.91%	81.26%
5	346	2231	70.01%	79.55%

Information filtering Experimental system also used the evaluation criteria given by Mobasher. Including recall rate, and precision rate, witch the recall rate is the content of the recommended items in the user accounts for users interested in all items of interest to the percentage accuracy rate is the recommended content items of interest to users total percentage of all the items recommended. RP measurement methods: the recall rate threshold is set from 0.1 to 1.0. Recall rates were calculated in 10 different threshold points, the system recommends documents the exact values. The results of our method and GA method are shown in Figure 1.

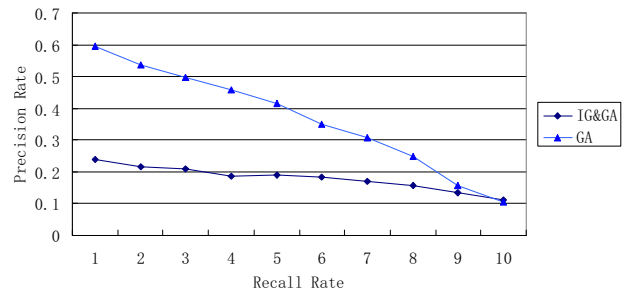


Figure 1. The results of our method and GA method

As can be seen from the results, based on the improvement of the IG and Genetic Algorithm feature selection algorithm for text effectively ensure the reduction of text classification and accuracy.

VI. CONCLUSIONS

Network information filtering technology which based on the user's information needing can meet the needs of users, so users can get information whatever they want quickly. This article based on the characteristics of text classification, for the lack of mutual information in feature selection, proposed a text feature selection method based on the improved information gain and genetic algorithm, improved the classification accuracy of the system in the large extent, this method is an effective feature selection method. In the future, we'll improve the feature selection methods and use the variety classification algorithms to compare their performance.

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